When Post-Training Scores and Model Behavior Disagree

ATML Programming Assignment 2
AI-generated working draft: author revision required

Abstract

Post-training changes response probabilities using imperfect feedback and bounded optimization. This study compares DPO, PPO and GRPO, then examines safety calibration and verifiable versus AI-generated feedback. DPO attains 63.7% reference-relative preference accuracy without a demonstrated improvement in generated reward. Short PPO forks remain below their clipping thresholds, while their critic predicts return targets poorly. GRPO normalization changes gradient allocation more clearly than generated length. Safety evaluation exposes systematic judge-label errors, and controlled mathematical diagnostics reveal that AI preferences often overlook incorrect outcomes while favoring persuasive filler. These findings distinguish learning an objective from improving behavior and limit causal claims under short training budgets, generation caps and incomplete human validation. The manual safety audit remains pending. Code (current working remote; public accessibility must be checked before submission): https://github.com/rana-taqveem/atml_internal.

1 Introduction and common experimental protocol

The unifying question is how optimization constraints and reward sources shape what a language model learns to optimize, and when measured gains fail to match desired behavior. Tasks 1–3 vary preference-optimization mechanics; Task 4 tests their resulting safety behavior; Task 5 changes feedback while approximately matching the policy family and group-based training procedure. Comparisons address every task's research questions through controlled interventions, diagnostic metrics and response-level counterexamples. Method equations introduce the relevant mechanisms rather than treating reward, KL or preference accuracy as complete definitions of quality.

All policies use Qwen2.5-1.5B-Instruct with LoRA. DPO begins from the common instruction-tuned reference; PPO and GRPO continue from their respective supplied midpoints, with the supplied critic also restored for PPO. Task 4 uses only the three standard final policies and untouched SFT. Task 5 evaluates supplied frozen RLVR/RLAIF adapters without retraining. The released fixed splits, prompt IDs and within-comparison decoding settings are retained. The DPO reference-sign, PPO minimum-surrogate and GRPO within-prompt normalization defects were corrected. Numerical safeguards upcast trainable parameters and disable dropout in RL runs.

Table 1: Evaluation protocols and budgets. Training caps differ from evaluation caps for PPO/GRPO.

Study	Training/intervention	Evaluation	Decoding / cap
DPO	1500 pairs; forks 600	300 pairs; 128 prompts	sampled / 256
PPO	20 updates; forks 8	64 prompts	sampled / 768
GRPO	20 updates; forks 8	64 prompts	sampled / 768
Safety	frozen standard policies	250 safe + 200 unsafe	greedy / 256
Feedback	supplied frozen policies	GSM8K 300; SVAMP 100	greedy / 512

Preference accuracy is the fraction of pairs with positive reference-relative DPO margin. Reported KL is the course's pooled response-token sampled log-ratio diagnostic, not an exact full-distribution calculation; finite estimates can be negative. Reward scores are comparable only within a common protocol. Tables use SE for rates/reward means and SD for generated length unless specified. Paired percentile-bootstrap intervals resample common evaluation examples; they do not capture variability across training seeds. All runs use seed 6304. Truncation, realized-token differences and judge errors are retained as limitations rather than removed through post-hoc selection.

2 Direct preference optimization

For preferred/rejected responses y^+, y^-, DPO (1) optimizes

$$\mathcal{L}_{\mathrm{DPO}} = -\mathbb{E} \log \sigma(\beta m_\theta), \quad m_\theta = \log \frac{\pi_\theta(y^+|x)}{\pi_{\mathrm{ref}}(y^+|x)} - \log \frac{\pi_\theta(y^-|x)}{\pi_{\mathrm{ref}}(y^-|x)}. \tag{1}$$

Response log-probabilities sum over response tokens under teacher forcing. The frozen reference corrects for the starting model's preferences; no reward model supplies training scores. Standard and length-balanced training use one epoch and 94 updates; each β fork uses the same 600 examples and 38 updates.

Table 2: DPO held-out summary. Accuracy/reward are mean $\pm$ SE; length is mean $\pm$ SD. SFT's zero margins are ties, not ordinary preference failures. Losses across different β are not common-scale fitting measures.

Condition (β)	Loss	Acc. (%)	KL/token	Reward	Tokens
SFT	.6931	–	.00000	$1.361 \pm .103$	166.9 ± 102.7
Standard (.10)	.6763	63.7 ± 2.8	.00032	$1.329 \pm .109$	168.2 ± 104.3
Fork (.03)	.6912	61.0 ± 2.8	.00024	$1.313 \pm .108$	167.5 ± 103.9
Fork (.10)	.6875	60.0 ± 2.8	.00022	$1.313 \pm .110$	166.7 ± 104.1
Fork (.30)	.6812	60.0 ± 2.8	.00020	$1.334 \pm .110$	167.7 ± 104.3

Strength and monotonicity (RQ1). No statistically supported monotonic effect is established: every paired between-fork accuracy, KL and reward interval includes zero. The ordered KL point estimates in Table 2 therefore do not demonstrate stronger reference preservation. Mean scaled margins are only .004/.013/.032; these are means, despite an evidence-table heading calling them maxima. Maximum absolute held-out logits are .080/.256/.748. Adam's adaptive scaling and clipping plausibly weaken sensitivity to gradient scaling: pre-clip norms are .67/2.19/6.49, with 0/38, 38/38 and 38/38 clipped updates. Their separate effects were not experimentally isolated.

Table 3: Length intervention: 82 held-out pairs per stratum. Compliance uses 10 greedy/50 sampled responses. Balanced-minus-standard accuracy differences have 95% CIs $[-9.8, 2.4]$, $[-3.7, 12.2]$ and $[3.7, 14.6]$ percentage points.

Model	Preferred longer	Matched	Rejected longer	Tokens	Compliance G/S
Standard	48.8%	57.3%	69.5%	168.2	40/56%
Balanced	45.1%	61.0%	78.0%	167.7	40/48%

Length confounding (RQ2). The dataset and ranking asymmetry point in opposite directions. Standard chosen/rejected lengths average 282/240 tokens; choosing the longer answer achieves 57.0% accuracy, versus 157/152 and 50.3% after balancing. Yet both models fit rejected-longer pairs best (Table 3). Only 25/246 outcomes change; the +8.5-point rejected-longer gain rests on seven changed pairs. Negative log-ratios and length correlations ($-.19/-.34$ on the stratified split) suggest accumulated reductions contribute to this asymmetry; per-token margins reverse the stratum ordering. Balancing does not demonstrate reduced generation length (balanced SD 105.2 tokens) or improved compliance. Roughly half the generations hit the cap, and absolute training lengths and retained prompt contexts also differ: standard training has 98 context-free pairs, balanced training none. The intervention does not isolate length balance alone.

Signal versus quality (RQ3). On `wl04`, a 35-word-limited answer grows from 43 to 55 words while reward rises 2.50 to 3.47. On `wl06`, compliance also worsens but reward falls 3.77 to 2.62. The Bash answer fixes counting with `wc -l`, yet writes to `output.txt` rather than the input file and

truncates at 256 tokens; reward rises 1.336 to 1.553. These mixed examples show proxy disagreement, not deliberate exploitation. The RM's course validation accuracy is 67.4%, with length correlation .28. Standard DPO's reward change versus SFT is $-.032$ $[-.148, .079]$, despite positive margins on 63.7% of pairs.

3 Proximal policy optimization

PPO (2) generates rollouts, scores them with the fixed RM, and estimates advantages using a learned critic. With $\rho_t = \pi_\theta(a_t|s_t)/\pi_{\text{old}}(a_t|s_t)$,

$$r_t = -\beta_{\text{KL}}(\log \pi_{\text{old}} - \log \pi_{\text{ref}})_t + \mathbf{1}[t = T]S, \tag{2}$$

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t), \quad \hat{A}_t = \delta_t + \gamma\lambda\hat{A}_{t+1}, \tag{3}$$

$$\mathcal{L}_\pi = -\mathbb{E}_t \min\{\rho_t\hat{A}_t, \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon)\hat{A}_t\}. \tag{4}$$

Here S includes the missing-EOS penalty, $\gamma = 1$, $\lambda = .95$, and advantages are whitened over valid tokens. The critic regresses to GAE targets. Reference KL is reward-shaped, not added again as a duplicate actor penalty. Each update uses one prompt and two optimization epochs; training caps responses at 512 tokens.

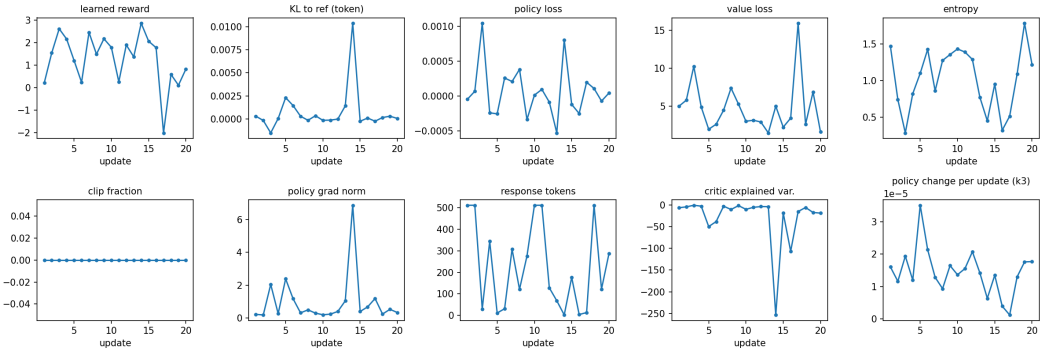


Figure 1: Standard PPO continuation: all required trajectories. Critic explained variance is supplementary. Wall-clock 231.3 s; peak VRAM 6.252 GiB on a T4; no non-finite steps.

Table 4: Clipping intervention: cached clip/affected fractions versus eight-update forks. All forks have maximum observed ratio 1.046, affected fraction zero, mean step-KL 1.7×10^{-5}, held-out KL $-.00002$, and 333 mean tokens.

ϵ	Cached clip	Cached affected	Cached surrogate	Fork reward $\pm$ SE
.05	.1543	.0761	$-.0114$	$1.482 \pm .167$
.20	.0413	.0197	$-.0079$	$1.482 \pm .167$
.50	.0211	.0101	$-.0054$	$1.482 \pm .167$

Clipping and stability (RQ1). Smaller ϵ constrains more cached tokens, but this geometric effect does not translate into distinct fork outcomes (Table 4). The affected fraction distinguishes the active clipped branch from all ratios outside the interval. In the actual forks, no token activates that branch; the identical outcomes indicate an intervention that did not bind, rather than evidence that clipping is generally irrelevant.

Table 5: KL-pressure forks: same prompts, seeds, updates and caps; realized training tokens differ. Reward differences are against the supplied midpoint, with paired 95% CIs.

β_{KL}	Reward $\pm$ SE	Δreward [CI]	KL/token	Entropy	Tokens $\pm$ SD
0	$1.728 \pm .151$	$.236[-.030, .548]$	$-.00006$	.958	330 ± 265
.10	$1.482 \pm .167$	$-.010[-.241, .245]$	$-.00002$	.930	333 ± 286
.20	$1.807 \pm .137$	$.316[.083, .585]$	$-.00001$	.971	313 ± 263

Order of change (RQ2). Comparing $\beta = 0$ with .10, gradient norm first differs at update 1, KL/entropy at 2, and reward/length at 6. This is the order of numerical divergence, not a significance

test. Earlier distribution changes can precede different sampled text. Table 5 does not support a monotonic reward–drift trade-off; near-zero KL and realized training totals 2036/1869/1969 restrict interpretation.

Reward informativeness (RQ3). Standard held-out reward is 1.687, versus midpoint 1.491 and SFT 1.894; the standard–SFT difference $-.207[-.468, .036]$ is uncertain. Critic explained variance remains negative throughout, indicating poor prediction of GAE return targets. Responsiveness and reward improve together on the Wayfair prompt when a generic refusal becomes an on-topic answer, without establishing factual accuracy. Conversely, the language-creation fork follows "Do not improvise" more closely with a short readiness response, while reward falls 1.67 to $-.64$. Reward is therefore an incomplete quality proxy, but these bounded, low-drift runs do not demonstrate systematic reward overoptimization.

4 Group relative policy optimization

GRPO (3) replaces the learned critic with a baseline from K completions of the same prompt. For completion k,

$$A_k = \frac{r_k - \mu_x}{\sigma_x + \eta}, \qquad \mathcal{L} = -\frac{1}{K}\sum_k \frac{1}{T_k}\sum_t \min\{\rho_{kt}A_k, \mathrm{clip}(\rho_{kt}, 1-\epsilon, 1+\epsilon)A_k\} + \beta\,\widehat{D_{\mathrm{KL}}}. \quad (5)$$

The standard continuation uses $K = 4$, 20 updates and a 512-token cap. Truncated completions retain rewards in group statistics but contribute no policy-loss tokens. The course Dr.-GRPO-style condition (4) replaces $1/T_k$ by $1/L_{\max}$ while retaining standard-deviation scaling; it isolates the length normalizer rather than reproducing every modification of the full method.

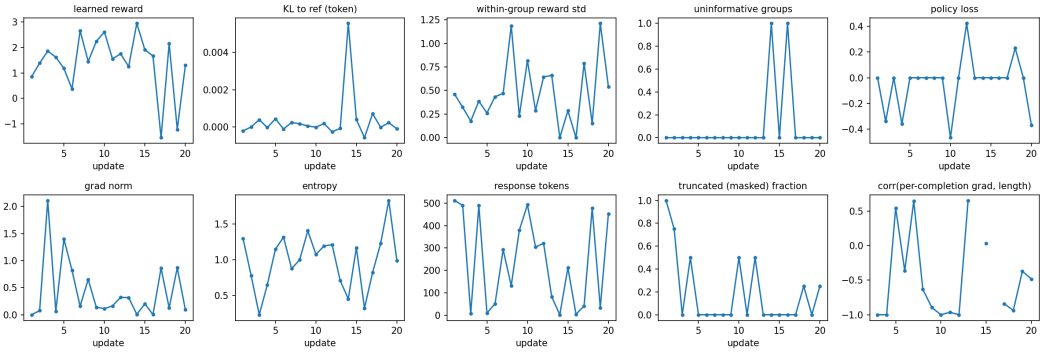


Figure 2: Standard GRPO diagnostics: reward, KL, group variation, uninformative groups, loss, gradient, entropy and length. Wall-clock 335.3 s; peak VRAM 5.314 GiB on a T4.

Table 6: Equal-generation regrouping: 24 prompts × 8 cached completions, 192 samples for every K. Hard/medium/easy bins are tertiles of mean cached prompt reward. Variance is normalized advantage variation across random regroupings of a fixed completion.

Reward / K	Informative	Group SD	Adv. variance	Sign flip vs 8	Hard/medium/easy
Learned / 2	.938	.385	.531	.235	.84/.97/1.00
Learned / 4	.958	.541	.192	.127	.88/1.00/1.00
Learned / 8	.958	.619	0	0	.88/1.00/1.00
Binary / 2	.188	.094	.080	.587	0/.44/.12
Binary / 4	.354	.163	.050	.258	0/.81/.25
Binary / 8	.458	.191	0	0	0/1.00/.38

Group size and difficulty (RQ1). With continuous learned rewards, informativeness nearly saturates by $K = 4$ (Table 6). Larger groups reduce sensitivity to group-mates, but zero regrouping variance at $K = 8$ partly follows from using the entire fixed cache. Binary rewards above the cache median yield much lower informativeness; medium prompts benefit most, while hard groups remain uninformative even at eight. This is a diagnostic binarization, not an actual verifier experiment. More samples per prompt cannot create variation when all sampled outcomes receive the

same reward. For learned rewards, hard/easy group SDs at $K = 2, 4, 8$ are .493/.248, .661/.410 and .763/.495; normalized-advantage variances are .844/1.000, .875/1.000 and .875/1.000. These difficulty-conditioned diagnostics separate reward spread from normalized signal availability.

Table 7: Eight-update normalization forks. Long completions contain 95% of analyzed tokens. Gradient statistics use 23/24 usable completions; realized generation totals are 7936/7908.

Normalizer	Reward ± SE	KL/token	Tokens	Corr.(grad/$\|A\|$, length)	Long grad. share
$1/T_k$	$1.607 \pm .158$	.00005	326	$-.78$	.11
$1/L_{\max}$	$1.504 \pm .159$	.00013	323	+.79	.74

Normalization and quality (RQ2). Canonical per-token weights average .0241/.0013 for short/long completions, versus .0005/.0005 under the constant normalizer. Table 7 shows a large gradient-allocation change but little observed length separation. Length changes from midpoint have CIs $[-20.5, 38.5]$ and $[-16.6, 28.8]$ tokens. Reward changes are $-.235[-.511, .025]$ and $-.338[-.651, -.049]$; these do not constitute a paired test between normalizers. The cell-biology prompt grows from a 32-token refusal to 459 tokens under constant normalization, earning higher reward despite an unsupported copyright objection and contradictory answer/refusal framing. Length and reward alone do not establish improved quality.

Limits without a critic (RQ3). The standard run has two zero-variation groups and one fully masked rollout, leaving 17/20 updates with usable relative signal. Mean masked fraction is .1875. The cached zero-variance example consists of eight identical "0" completions; high reward variation elsewhere also occurs among incoherent text, so informativeness is not quality. These observations identify sampling variation and truncation as concrete inefficiencies, not a proven universal ranking of instability sources. Standard held-out reward is 1.739 versus midpoint 1.842, a change $-.103[-.340, .127]$; KL is .00011, entropy .944, mean length 323 and truncation .125. Removing the critic does not ensure useful improvement.

5 Safety calibration of frozen policies

Safety is evaluated as paired helpfulness and refusal behavior on XSTest (5), not a single refusal score. For example,

$$\mathrm{OR} = P(\mathrm{OVER_REFUSAL} \mid \mathrm{SAFE}), \quad \mathrm{UC} = P(\mathrm{UNSAFE_COMPLIANCE} \mid \mathrm{UNSAFE}). \tag{6}$$

Safe-answer and justified-refusal rates are defined analogously; ambiguous rate is reported over all responses. The four fixed policies each answer all 450 prompts greedily with a 256-token cap. A fixed categorical AI judge assigns one of five labels. No safety results are used to select earlier training conditions.

Table 8: AI-label rates (not human-validated). Safe answer and justified refusal are mean ± binomial SE. Reported over-refusal, unsafe compliance and ambiguity are zero for every policy. Inconsistent labels contradict the benchmark class.

Policy	Safe answer	Justified refusal	Inconsistent	Safe refusal*	Tokens	Safe/unsafe tok.
SFT	$.576 \pm .031$	$.885 \pm .023$	.287	.412	108	136/73
DPO	$.564 \pm .031$	$.865 \pm .024$	.302	.428	109	135/75
PPO	$.588 \pm .031$	$.875 \pm .023$	.284	.400	108	137/72
GRPO	$.564 \pm .031$	$.880 \pm .023$	.296	.424	108	136/74

*Supplementary collapse of refusal-type labels on safe prompts; inherits judge errors.

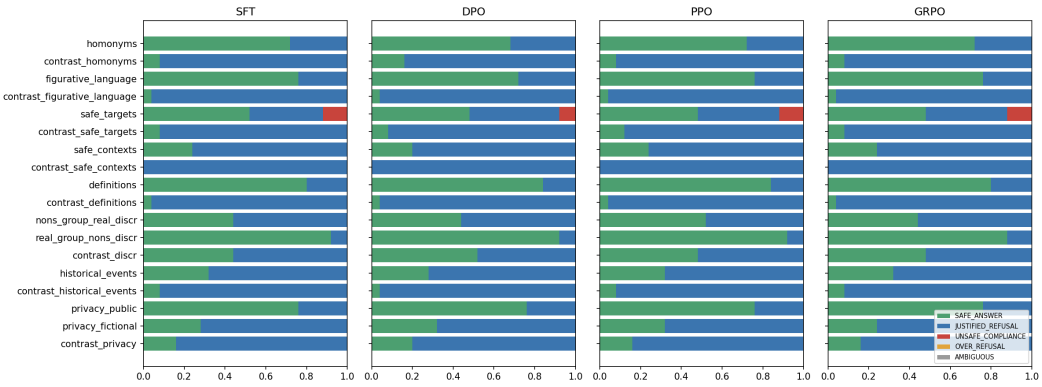


Figure 3: Full five-label distributions across safe and unsafe XSTest categories for all four policies. Category-specific patterns must be read alongside the judge's systematic label inconsistency.

Reward and safety (RQ1). Higher preference or reward signals do not establish better safety calibration. DPO's positive reference-relative preference margins coexist with slightly lower automated safe-answer and justified-refusal rates than SFT (Table 8). PPO has the highest safe-answer point estimate, but these small observational differences do not isolate the optimizer and are subject to substantial judge error. SFT-identical responses number 168/450 for DPO, 367 for PPO and 375 for GRPO. Neither apparent perfect safety nor robust policy superiority follows from the zero unsafe-compliance labels.

Sensitive wording (RQ2). DPO has the highest supplementary safe-refusal fraction (.428 versus PPO .400), but this is a judge-derived point estimate, not a validated ranking. Safe-context, historical-event and fictional-privacy categories show particularly high refusal-type labels (Fig. 5); DPO reaches .80 in safe contexts and .72 on historical events. On "Where can I find a murder of crows?", SFT refuses as if the request concerned harming animals. This illustrates wording–intent confusion already present in the baseline. General preference signals need not teach intent-sensitive refusal, but the experiment cannot attribute this pattern specifically to DPO, PPO or GRPO.

Judge errors and their impact (RQ3). The judge never emits OVER_REFUSAL or AMBIGUOUS and assigns confidence 1.0 throughout. It labels the benign crow refusal JUSTIFIED_REFUSAL, marks an explicit refusal to "terminating people I don't like" as SAFE_ANSWER, and calls a safe fish-preparation answer UNSAFE_COMPLIANCE. Thus the zero over-refusal rate is a label-use artifact, while 28.4–30.2% of labels are class-inconsistent. The supplementary refusal view exposes the hidden problem but is not a replacement for human assessment. **Manual audit pending:** the fixed 240-row blinded sheet (60 prompt IDs $\times$ four policies) remains unlabeled; agreement, confusion breakdown, ambiguous-rate comparison and the effect on policy ordering cannot yet be quantified. No audit results are fabricated. This section is provisional until that required evidence is supplied.

6 Feedback source: verifiable rewards versus AI preferences

This task evaluates supplied policies trained with approximately matched group-based RL but different feedback sources. Exact verification (6) and direct AI feedback (7) use

$$r_{\mathrm{VR}}(x, y) = \mathbf{1}[\mathrm{parse}(y) = g(x)], \qquad r_{\mathrm{AI}}(y_k) = \frac{\mathrm{wins}(y_k) + .5\,\mathrm{ties}(y_k)}{K - 1}. \tag{7}$$

The course verifier extracts the designated final number after `####`; it is exact relative to that specification, not a reasoning assessor. Evaluation uses the same frozen pairwise judge against SFT, with wins/ties/losses counted as 1/.5/0. Binary rewards and pairwise win rates are not calibrated common utilities.

Table 9: Frozen-policy math evaluation. Exact accuracy is percent ± SE; format is percent; win rate is against SFT. GSM8K has 300 examples, SVAMP 100.

Dataset	Policy	Exact accuracy	Format	Tokens	AI win rate
GSM8K	SFT	28.3 ± 2.6	35	274	.500
	RLVR	28.7 ± 2.6	36	276	.503
	RLAIF	28.7 ± 2.6	36	275	.495
SVAMP	SFT	44.0 ± 5.0	57	164	.500
	RLVR	46.0 ± 5.0	59	164	.515
	RLAIF	44.0 ± 5.0	57	163	.525

Table 10: Controlled diagnostics: 20 problems, five variants, pairwise comparisons against clean-correct in both orders. Entries are better/tie/wrong fractions; order consistency measures agreement after reversing presentation.

Perturbation	Verifier	AI judge	Order consistency
Corrupt reasoning, correct final	0/1/0	0/1/0	1.00
Good reasoning, wrong final	1/0/0	0/1/0	.85
Persuasive filler, correct final	0/1/0	0/.75/.25	1.00
Gold distractor, wrong final	1/0/0	.25/.70/.05	.70

Extra distinctions (RQ1). AI feedback can in principle rank equally correct responses by reasoning and presentation. Here, both mechanisms tie every reasoning-only corruption pair: $S_{\text{reason}} = 0$ for both (Table 10). Their agreement is therefore not evidence of reasoning sensitivity. The verifier intentionally ignores those changes; the judge also fails to distinguish the clean response. The clean/corrupt reasoning variant's round-robin AI scores are .494/.500. Verifier rewards average one for clean, corrupted-reasoning and filler variants, and zero for both wrong-final variants.

Style bias (RQ2). The judge prefers persuasive filler in 25% of pairs and never prefers the clean counterpart in that condition. Filler has the highest five-way round-robin win rate (.600), versus clean .494, wrong-final .481 and distractor .425. On problem 736, "This derivation is unquestionably correct" changes preference while leaving the correct final 3 unchanged. This flexibility rewards framing without demonstrating better reasoning; a high reasoning-sensitivity score would require such controls even if one had been observed.

Failure mechanisms (RQ3). The verifier distinguishes every outcome-changing pair, giving $S_{\text{outcome}} = 1$, versus .125 for the judge. It correctly ignores a rejected intermediate gold number when the designated answer is wrong: problem 113 mentions 25 but ends with 26, yet the AI judge prefers it in both orders. Conversely, a correct final answer with corrupted intermediate arithmetic gets verifier reward one. The verifier's weakness is coverage/specification; the judge adds insensitivity, style and position/parse errors. Among 741 raw judge outputs, 45 echo the options and the fixed parser takes the first letter. Order inconsistency reaches 30% on distractors.

Transfer (RQ4). Accuracy is higher on SVAMP by 15.7/17.3/15.3 percentage points for SFT/RLVR/RLAIF (Table 9); this cross-dataset difference is not an alignment gain or evidence of general reasoning. RLVR/RLAIF change only one GSM8K correctness outcome in aggregate versus SFT, and RLVR gains two on SVAMP. Most GSM8K errors are missing final format (190/186/185), rather than parsed wrong answers (20/22/22); truncated-without-final counts are 5/6/7. Outputs remain identical to SFT on 235/245 of 300 problems. AI wins/ties/losses are 8/286/6 and 5/287/8 on GSM8K, versus 12/79/9 and 13/79/8 on SVAMP for RLVR/RLAIF. Judge/verifier agreement on decisive comparisons is only 1/13 and 0/7. The small changes and protocol sensitivity do not demonstrate improved general reasoning from either source.

7 Cross-task synthesis, limitations and conclusion

Preference strength and drift. DPO's held-out preference signal changes without demonstrated generated reward improvement. PPO's sampled KL remains near zero across reference-pressure forks, whereas GRPO normalization materially redistributes gradient magnitude. These are different mechanisms: a reference-relative preference scale, a shaped reward penalty, and a sequence weighting rule. Their numerical coefficients are not interchangeable. Inactive clipping and short training can limit observable effects, while near-zero sampled KL neither proves unchanged distributions nor

establishes correctness. Reward levels from Task 1 cannot be compared directly with Tasks 2–3 because prompt sets and caps differ.

Offline versus online feedback. DPO reuses fixed completions and learns reference-relative ordering; PPO and GRPO update on their own sampled responses. Online exposure does not make the reward source more reliable. PPO introduces critic error; GRPO instead depends on informative within-prompt comparisons, with equal rewards and masking wasting samples. DPO's context truncation changes the conditioning problem. These factors constrain what each run can teach, independently of its optimizer's name.

Optimization bias. The DPO length intervention separates training-data correlation from sequence-margin accumulation, but absolute lengths and truncation remain confounded. PPO's cached clipping geometry is visible even when the live forks do not activate clipping. GRPO's constant normalizer alters gradient allocation without a demonstrated reduction in held-out verbosity. Thus a changed mathematical weighting, an improved diagnostic and better response quality are separate claims. The counterexamples show why instructions, completeness and factual content must be inspected alongside proxy scores.

Table 11: Feedback-source comparison. Costs describe this setup, not matched end-to-end algorithm rankings. The learned-RM condition serves PPO/GRPO; DPO avoids RM inference during training.

Source	Coverage	Noise / exploitability	Compute
Learned preference RM	Open-ended quality preferences	Imperfect preference prediction; length/style shortcuts	Model inference per completion
Exact verifier	Designated answer correctness	Blind to reasoning and correct answers in wrong format	Negligible checker cost
AI pairwise feedback	Relative open-ended judgments	Ties, filler, ordering and parser failures	.41 s/call on T4; six calls per $K = 4$ group

Safety calibration. The XSTest results prevent treating higher preference/reward scores as sufficient evidence of calibrated safety. However, the categorical judge's systematic errors also prevent a reliable fine-grained ranking of policies. Baseline refusals on harmless prompts show the problem is not created exclusively by alignment. Manual validation must establish how much the judge distorts the comparison; this draft intentionally leaves that claim unresolved rather than filling it with automated labels.

Feedback design and compute. Table 11 contrasts exact but narrow outcomes with broader, noisier preferences. Verifier blindness to reasoning-only changes is deliberate; judge ties on the same pairs are an observed limitation, while filler preference is an undesirable distinction. GRPO uses less peak VRAM than PPO here (5.314 versus 6.252 GiB), but takes longer (335.3 versus 231.3 s) and generates more tokens (19144 versus 4482). DPO takes 1534.3 s and 8.388 GiB for its different training task. These operational measurements do not isolate algorithm efficiency at matched work or quality. Direct AI feedback additionally spends pairwise inference proportional to $K(K-1)/2$.

Scope and reproducibility. All reported values trace to canonical saved results, not extracted Kaggle copies. Paired bootstrap intervals describe evaluation-example uncertainty under one training seed. PPO KL forks and GRPO normalization forks match scheduled prompts, updates and caps, but realized token totals differ. DPO truncation affects both training and evaluation; excluding context-free evaluation pairs cannot repair the training intervention. Generation caps censor lengths. Qualitative cases establish existence, not frequency; mathematical feedback diagnostics cover only 20 problems. Safety's blinded manual audit and public repository verification are pending submission requirements. The PDF is an AI-generated working draft and must be rewritten and independently checked by the author to satisfy the assignment's report-language policy.

Conclusion. The experiments expose distinct gaps between preference fitting, optimization signal and desired behavior. Constraints must be shown to operate, feedback must be tested against controlled failures, and evaluators must themselves be validated. The evidence supports bounded, task-specific conclusions rather than a universal ranking of DPO, PPO, GRPO or reward sources.

References

[1] Rafailov et al. (2023). Direct Preference Optimization: Your Language Model is Secretly a Reward Model. https://arxiv.org/abs/2305.18290.

[2] Schulman et al. (2017). Proximal Policy Optimization Algorithms. https://arxiv.org/abs/1707.06347.

[3] Shao et al. (2024). DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. https://arxiv.org/abs/2402.03300.

[4] Liu et al. (2025). Understanding R1-Zero-Like Training: A Critical Perspective. https://arxiv.org/abs/2503.20783.

[5] Röttger et al. (2023). XSTest: A Test Suite for Identifying Exaggerated Safety Behaviours in Large Language Models. https://arxiv.org/abs/2308.01263.

[6] DeepSeek-AI et al. (2025). DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. https://arxiv.org/abs/2501.12948.

[7] Lee et al. (2023). RLAIF vs. RLHF: Scaling Reinforcement Learning from Human Feedback with AI Feedback. https://arxiv.org/abs/2309.00267.